Increasing Productive BAYMAX Examples to Two Hundred

Experiment index

Question and Context

Could more domain-specific productive examples recover task completion without adding more clarification/refusal rows?

Change

The historical ACT-balanced mixture selected 400 BAYMAX rows: 200 ACT, 100 CLARIFY and 100 REFUSE. The remaining 600 raw slots came from xLAM.

This is the old ACT-balanced model, not generator14 ACT200 or the cleaned anchored checkpoint. The similar class counts do not make the data revisions identical.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Historical ACT200 / legacy metrics	150	47/150 (0.313)	0.553	0.805	0.025	0.086
Historical ACT200 / decision metrics	150	47/150 (0.313)	0.553	0.805	0.025	0.086
Historical ACT200 / legacy metrics	30	6/30 (0.200)	0.533	0.828	0.200	0.100
Historical ACT200 / decision metrics	30	6/30 (0.200)	0.533	0.828	0.200	0.100
Historical ACT200 / strict replay	150	46/150 (0.307)	0.493	0.738	0.000	0.086
Historical ACT200 / strict replay	30	6/30 (0.200)	0.500	0.794	0.200	0.100

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
Historical ACT200 / decision metrics (150 cases)	0.387	0.000

Evaluation	Macro-F1	Invalid-output rate
Historical ACT200 / decision metrics (30 cases)	0.397	0.000
Historical ACT200 / strict replay (150 cases)	0.349	0.120
Historical ACT200 / strict replay (30 cases)	0.368	0.067

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Result Provenance

- Historical ACT200 / legacy metrics: 2026-09-27 UTC, 150 cases.
- Historical ACT200 / decision metrics: 2026-09-27 UTC, 150 cases.
- Historical ACT200 / legacy metrics: 2026-09-27 UTC, 30 cases.
- Historical ACT200 / decision metrics: 2026-09-27 UTC, 30 cases.
- Historical ACT200 / strict replay: 2026-09-28 UTC, 150 cases.
- Historical ACT200 / strict replay: 2026-09-28 UTC, 30 cases.

Interpretation

The historical evaluation recorded 47/150 successes; strict replay recorded 46/150. Template-shift success remained 6/30.

The increase in productive exposure improved some established-suite outcomes but also increased premature actions. It was not a complete improvement in decision policy.

Limits

The "decision-metrics" evaluation added explicit decision reporting but did not create a new trained model. The "legacy-metrics" file remains preserved.

Strict replay fixes measurement only, not the old training partitions or seed initialization. Later ACT200 runs are not single-variable comparisons with this checkpoint.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.